

The regular style

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1. Introduction

This is the standard. If you're not going to pick anything else, you're getting this. Don't worry, I don't think you're boring for using this style. You have three outfits with statement pieces in them, but not this style. This is jeans, a t-shirt, a flannel, and some sneakers—timeless, comfortable, and effortless, but not high-fashion. You're not going to be turning heads, but you're going to be very presentable, affable, and capable of dealing with most temperatures. It's *Balatro*, a game that always works when you don't feel like playing anything in particular.

I chose STIX and Inter, because both of them are, to me, the “new” version of Times and Helvetica, respectively—they're workhorses, cockroaches that have survived the decades because they just work. They're not particularly bold fonts, but they're quite legible and capable of efficiently delivering text to readers. Both fonts are good on screens and in print—even if Inter is, for all intents and purposes a screen font. (Matthew Butterick, if you're reading this, please don't sue me for using a screen font.) STIX is good at both text and math, so it's quite suitable for mathematical and scientific documents, but it's no slouch for straight text, neither. Both of these fonts have a well made italic shape and boldface.

You can certainly let your hair down a little bit with this style, but it's also not going to be the best for very informal communications. Please don't make a rave poster or a birthday party invite using this style—do make a poster for an *underground* rave with this, since it looks like a regular document and therefore most people won't give it a second thought.

2. Standard demo

The following text is taken from Lieb, E., Schultz, T., Mattis, D. (2004).¹ Please, go read the actual paper, it's great.

¹Two Soluble Models of an Antiferromagnetic Chain. In: Nachtergaele, B., Solovej, J.P., Yngvason, J. (eds) Condensed Matter Physics and Exactly Soluble Models. Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-662-06390-3_35

2.1. Formulation

The first model consists of N spin $1/2$'s (N even) arranged in a row and having only the nearest neighbor interactions. It is

$$H_\gamma = \sum_i [(1 + \gamma) S_i^x S_{i+1}^x + (1 - \gamma) S_i^y S_{i+1}^y],$$

where γ is a parameter characterizing the usual degree of anisotropy in the xy -plane. Because the Hamiltonian only involves the x - and y - components of the spin operators, we call this model the XY model.

To solve the XY model, we first introduce the raising and lowering operators

$$a_i^\dagger = S_i^x + i S_i^y \text{ and } a_i = S_i^x - i S_i^y$$

in terms of which the Pauli spin operators are

$$S_i^x = (a_i^\dagger + a_i)/2; S_i^y = (a_i^\dagger - a_i)/2i; S_i^z = a_i^\dagger a_i - \frac{1}{2}$$

and the Hamiltonian is

$$H_\gamma = \frac{1}{2} \sum_i [(a_i^\dagger a_{i+1} + \gamma a_i^\dagger a_{i+1}^\dagger) + \text{h.c.}].$$

The ground state Ψ_0 is the state with no elementary excitations:

$$\eta_k \Psi_0 = 0, \text{ all } k.$$

The ground-state energy, according to (A-12) is

$$E_0 = -\frac{1}{2} \sum_k \Lambda_k.$$

In the limit $N \rightarrow \infty$, the sum can be replaced by an integral giving

$$\begin{aligned} \frac{E_0}{N} &= -\frac{1}{4\pi} \int_{-\pi}^{+\pi} [1 - (1 - \gamma^2) \sin^2 k]^{1/2} dk \\ &= -\frac{1}{\pi} \mathcal{E}(1 - \gamma^2), \end{aligned}$$

where $\mathcal{E}(k^2)$ is one of the complete elliptic integrals (13).